

Maxwell eqs $\Longleftrightarrow$ 物质方程　　波动方程

$$\begin{cases} \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \\ \nabla \times \vec{H} = \frac{\partial \vec{D}}{\partial t} \\ \nabla \cdot \vec{D} = 0 \\ \nabla \cdot \vec{B} = 0 \end{cases}$$

$$\vec{D} = \varepsilon_0 \vec{E} + \vec{P}$$

$\vec{P}$ Taylor 展开 $\varepsilon_0 (\underbrace{\chi^{(1)}\vec{E}}_{\text{线性}} + \underbrace{\chi^{(2)}:\vec{E}\vec{E}}_{\text{非线性}\approx 0} + \chi^{(3)}:\vec{E}\vec{E}\vec{E} + \cdots)$

$$= \varepsilon_0 (1+\chi^{(1)})\vec{E}$$

$$= \varepsilon_0 \varepsilon_r \vec{E} = \varepsilon \vec{E}$$

?

ε

- 各向同性
  - 真空　$\vec{D} = \varepsilon_0 \vec{E}$
  - 均匀介质
    - $\vec{D} = \varepsilon \vec{E}$　无色散
    - $\vec{D}(\omega) = \varepsilon_\omega \vec{E}(\omega)$　有色散
    - $\vec{D} = (\varepsilon' + i\varepsilon'')\vec{E}$　吸收
  - 非均匀介质　$\vec{D} = \varepsilon(r)\vec{E}$
- 各向异性（晶体）　$\vec{D} = [\varepsilon]\vec{E}$

波动方程

① 真空.

$$\nabla\times(\nabla\times\vec{E}) = \nabla\times\left(-\frac{\partial\vec{B}}{\partial t}\right) = -\mu_0\frac{\partial}{\partial t}(\nabla\times\vec{H})$$

$$Left = \nabla\times(\nabla\times\vec{E}) = \nabla(\nabla\cdot\vec{E}) - \nabla^2\vec{E} = -\nabla^2\vec{E}$$

($\nabla\cdot\vec{E}=0$)

$$Right = -\mu_0\frac{\partial}{\partial t}\left(\frac{\partial}{\partial t}\vec{D}\right) = -\mu_0\varepsilon_0\frac{\partial^2}{\partial t^2}\vec{E} = -\frac{1}{c^2}\frac{\partial^2}{\partial t^2}\vec{E} \qquad c=\frac{1}{\sqrt{\mu_0\varepsilon_0}}$$

真空中波动方程

$$\nabla^2\vec{E} - \mu_0\varepsilon_0\frac{\partial^2}{\partial t^2}\vec{E} = \nabla^2\vec{E} - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\vec{E} = 0$$

无色散情形:

$$\nabla^2\vec{E} - \mu\varepsilon\frac{\partial^2}{\partial t^2}\vec{E} = \nabla^2\vec{E} - \frac{1}{v^2}\frac{\partial^2}{\partial t^2}\vec{E} = 0$$

有色散?

$$t \xleftarrow{\text{Fourier trans}} \omega$$

$$\vec{E}(t) \xrightarrow{FT} \int_{-\infty}^{\infty}\vec{E}_\omega(\vec{r},t)\,d\omega = \int_{-\infty}^{\infty}\vec{E}_\omega(\vec{r})\,e^{-i\omega t}\,d\omega$$

同理, 代入:

$$\nabla^2\vec{E}(\vec{r},t) - \frac{1}{v^2}\frac{\partial^2}{\partial t^2}\vec{E}(\vec{r},t) = 0$$

有:

$$\nabla^2\vec{E}(\vec{r}) + \frac{\omega^2}{v^2}\vec{E}(\vec{r}) = \nabla^2\vec{E}(\vec{r}) + k^2\vec{E}(\vec{r}) = 0 \qquad k=\frac{\omega}{v}=\frac{2\pi}{\lambda}$$

亥姆霍兹方程

$$\nabla^2\vec{E}(\vec{r}) + k^2\vec{E}(\vec{r}) = 0$$

$\frac{\partial^2}{\partial t^2} = -\omega^2$ ， $\frac{\partial}{\partial t} = -\omega$ ， $\nabla = i\vec{k}$ $\nabla^2 = -k^2$

非均匀介质 $\varepsilon = \varepsilon(r)$

$$\vec{D} = \varepsilon(r)\vec{E}$$

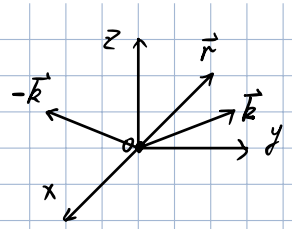
$$\nabla\times(\nabla\times\vec{E}) = \nabla(\nabla\cdot\vec{E}) - \nabla^2\vec{E}$$

$$= \nabla(\nabla\cdot\frac{\vec{D}}{\varepsilon}) - \nabla^2\frac{\vec{D}}{\varepsilon}$$

$\neq 0$

$$\nabla\cdot\vec{D} = \nabla\cdot(\varepsilon\vec{E}) = (\nabla\cdot\varepsilon)\vec{E} + \varepsilon\nabla\cdot\vec{E} = 0 \Longrightarrow \nabla\cdot\vec{E} = -\frac{\nabla\cdot(\varepsilon\vec{E})}{\varepsilon}$$

$$-\nabla(\frac{\nabla\cdot\varepsilon\vec{E}}{\varepsilon}) - \nabla^2 E = -\frac{1}{c^2}\frac{\partial^2}{\partial t^2}\vec{E}$$



平面波 $\vec{E}_r = \vec{E}_0 e^{i\vec{k}\cdot\vec{r}}$ 特解

代入亥姆霍兹 eg. 有:

$$\nabla^2(\vec{E}_0 e^{i\vec{k}\cdot\vec{r}}) + k^2(\vec{E}_0 e^{i\vec{k}\cdot\vec{r}}) = 0$$

$$\vec{E}_0(\nabla^2 e^{i\vec{k}\cdot\vec{r}} + k^2 e^{i\vec{k}\cdot\vec{r}}) = 0$$

$$\vec{E}(\vec{r},t) = \vec{E}(\vec{r})e^{-i\omega t} = \vec{E}_0 e^{-i(\omega t - \vec{k}\cdot\vec{r} + \varphi_0)}$$

时间 空间 初相位

或也可写为三角函数形式:

$$\vec{E}(\vec{r},t) = \vec{E}_0\cos(\omega t - \vec{k}\cdot\vec{r} + \varphi_0)$$

$$= \frac{1}{2}\vec{E}_0\left(e^{-i(\omega t - \vec{k}\cdot\vec{r} + \varphi_0)} + e^{i(\omega t - \vec{k}\cdot\vec{r} + \varphi_0)}\right)$$

- $\omega t - \vec{k}\cdot\vec{r} + \varphi_0 = const \Longrightarrow$ 等相面.
- 相速度 v_ϕ $\quad d(\omega t - \vec{k}\cdot\vec{r} + \varphi_0) = \omega dt - \vec{k}\cdot d\vec{r} = 0$

$$\Rightarrow \frac{\omega}{k} = \frac{d\vec{r}}{dt}$$

- 群速度 v_g

横波 or 纵波？

传播方向与振动方向 // 纵波

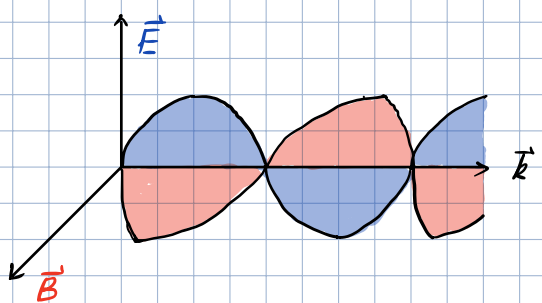
⊥ 横波.

① $\nabla\cdot\vec{D} = \varepsilon\nabla\vec{E} = \varepsilon i\vec{k}\cdot\vec{E} = 0$

② $\nabla\times\vec{E} = -\frac{\partial\vec{B}}{\partial t}$ $\quad i\vec{k}\times\vec{E} = i\omega\vec{B}$

$\vec{k}\cdot(\vec{k}\times\vec{E}) = i\omega\,\vec{k}\cdot\vec{B} = 0$

} $\vec{k}$、$\vec{E}$、$\vec{B}$ 两两垂直.



$$\frac{|\vec{E}|}{|\vec{B}|} = \frac{\omega}{k} = v \sim 10^8$$

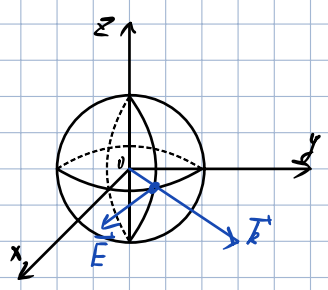
考虑球坐标：$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right)$ $\quad \frac{\partial}{\partial\theta} = \frac{\partial}{\partial\varphi} = 0$

$$\frac{\partial^2}{\partial r^2}[r\vec{E}(r)] + k^2[r\vec{E}(r)] = 0$$

$$\vec{E}(r) = \frac{\vec{E}_0}{r}e^{i\vec{k}\cdot\vec{r}}$$

球面波

$$\vec{E}(\vec{r},t) = \frac{\vec{E}_0}{r}e^{-i(\omega t - \vec{k}\cdot\vec{r} + \varphi_0)}$$



柱坐标系：

$$\nabla^2 = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial}{\partial r}\right)$$

$$\frac{\partial^2}{\partial r^2}\vec{E}(\vec{r}) + \frac{1}{r}\frac{\partial}{\partial r}\vec{E}(\vec{r}) + k^2\vec{E}(\vec{r}) = 0$$

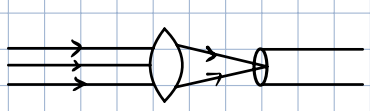
Bessel eq.

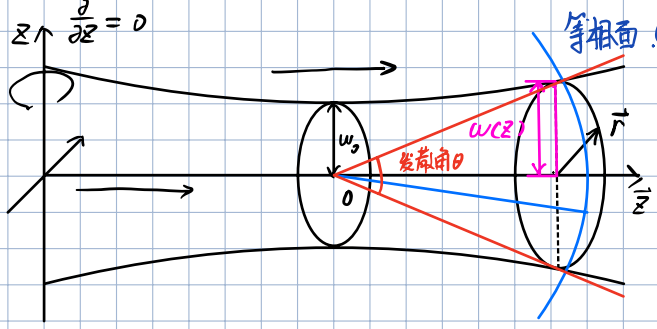
解为：

$$\vec{E}(\vec{r}) = \frac{\vec{E}_0}{\sqrt{r}} e^{-i(\omega t - \vec{k}\cdot\vec{r} + \varphi_0)}$$

高斯球面波.

e.x.t 激光.





等相面(球面波)曲率半径 $R(z)$

∞近似为平面波.

∇^2

$r\vee$

$z\vee$

φX

$$\frac{\partial^2\vec{E}}{\partial t^2}+\frac{1}{r}\frac{\partial\vec{E}}{\partial r}+\frac{\partial^2\vec{E}}{\partial z^2}+k^2\vec{E}=0$$

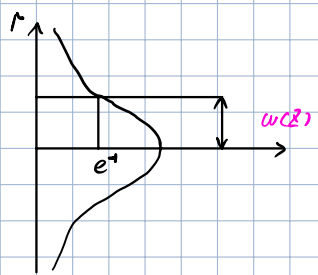
假设 $\vec{E}$ 有特解形式: $\vec{E}=Ae^{i\vec{k}\cdot\vec{r}}=Ae^{ikz}$

$$E(r,z)=\underbrace{\frac{A_0}{\omega(z)}}_{\text{振幅}}\exp\left(\frac{-r^2}{\omega^2(z)}\right)\exp\left\{i\left[kz+\underbrace{\frac{kr^2}{2R(z)}}_{\text{相位}}-\varphi(z)\right]\right\}$$

$\omega(z)$: 表高斯光束宽度　　$R(z)$: 等相面曲率半径.

$$\omega(z)=\omega_0\sqrt{1+\frac{z^2}{z_0^2}}$$

$$R(z)=z\left(1+\frac{z_0^2}{z^2}\right)$$



其中, $z_0=\frac{\pi\omega_0^2 n}{\lambda_0}=\frac{\pi\omega_0^2}{\lambda}$

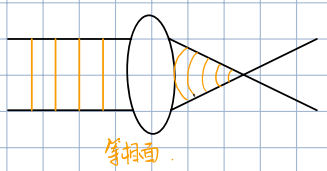
旁轴近似下: $R=\sqrt{r^2+z^2}\approx z\left(1+\frac{r^2}{2z^2}\right)=z+\frac{r^2}{2z}$

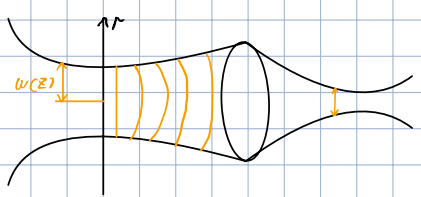
发散角. $\theta = \lim_{z\to\infty} \frac{2\omega(z)}{z} \quad \frac{2\lambda_0}{\pi\omega_0 n} \sim 10^{-5}$

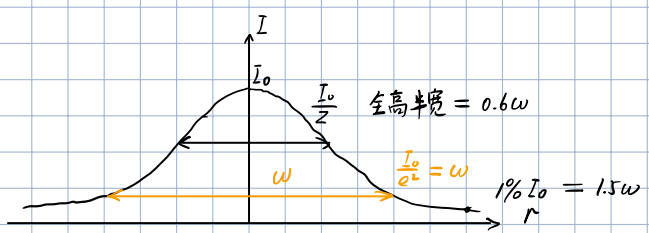
传输距离很远，光斑很大，功率密度很小

$$\nabla^2\vec{E} + k^2\vec{E} = 0$$

$$\vec{E} = \frac{A_0}{\omega(z)} \exp\left(-\frac{r^2}{\omega^2(z)}\right) \exp\left\{i\left[kz + \frac{kr^2}{2R(z)} - \varphi(z)\right]\right\}$$

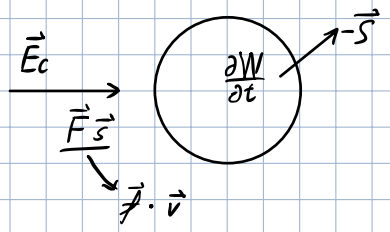






能量、能流.

$\vec{F}$: Lorentz force.



能量守恒：

$$\nabla\cdot\vec{S} = -\frac{\partial \vec{w}}{\partial t} - \vec{f}\cdot\vec{v}$$

prove.

Maxwell eqs.:

$$\begin{cases} \nabla\times\vec{E} = -\dfrac{\partial\vec{B}}{\partial t} & ① \\ \nabla\times\vec{H} = \vec{J} + \dfrac{\partial\vec{D}}{\partial t} & ② \\ \nabla\cdot\vec{D} = \rho \quad (\vec{D} = \varepsilon\vec{E}) & ③ \\ \nabla\cdot\vec{B} = 0 \quad (\vec{B} = \mu\vec{H}) & ④ \end{cases}$$

$$\vec{f}\cdot\vec{v} = (\rho\vec{E} + \vec{J}\times\vec{E})\cdot\vec{v} = \rho\vec{E}\cdot\vec{v} + (\vec{J}\times\vec{E})\cdot\vec{v}$$

($(\vec{J}\times\vec{E})\cdot\vec{v} = 0$)

$$= \vec{J}\cdot\vec{E}$$

$$\overset{②}{=\!=} (\nabla\times\vec{H} - \frac{\partial\vec{D}}{\partial t})\vec{E} = \vec{H}(\nabla\times\vec{E}) - \nabla(\vec{E}\times\vec{H}) - \vec{E}\frac{\partial\vec{D}}{\partial t}$$

$$= -\vec{H}\frac{\partial\vec{B}}{\partial t} - \nabla(\vec{E}\times\vec{H}) - \vec{E}\frac{\partial\vec{D}}{\partial t}$$

$$= -\frac{\partial w}{\partial t} - \nabla\vec{S}$$

$$\begin{cases} \vec{S} = \vec{E} \times \vec{H} \\ \int dw = \int \vec{E} d\vec{D} + \int \vec{H} d\vec{B} \\ \quad = \int \varepsilon \vec{E} d\vec{E} + \int \mu \vec{H} d\vec{H} \end{cases} \Longrightarrow w = \frac{1}{2}(\varepsilon E^2 + \mu H^2)$$

$$= \frac{1}{2}(\vec{E}\cdot\vec{D} + \vec{H}\cdot\vec{B})$$

* question: $\vec{S}$ 表能量传输方向，与 $\vec{k}$ 方向相同吗？

平面波 $\vec{E} = \vec{E}_0 e^{i(\omega t - \vec{k}\cdot\vec{r})}$

$\omega \sim 10^{15}$ 无法精细测，只能测 $\langle w \rangle = \int_0^T w dt$

$\nabla = i\vec{k}$ $\frac{\partial}{\partial t} = -i\omega$ 代入 Maxwel eqs.

$$\vec{k} \times \vec{E} = \omega \vec{B}$$

$$\frac{|E|}{|B|} = v = \frac{1}{\sqrt{\mu\varepsilon}}$$

$$w = \mu H^2 = \varepsilon E^2 = \frac{1}{\mu} B^2$$

Homework 1.2 1.6 1.7

2.1 2.10 2.14 next Friday

Maxwel eqs

物质 eq

Lorentz force
$\vec{f} = \rho\vec{E} + \vec{J}\times\vec{B}$

$$\Rightarrow \nabla\vec{S} = -\vec{f}\vec{v} - \frac{\partial w}{\partial t}$$

$$\begin{cases} \vec{S} = \vec{E}\times\vec{H} \\ dw = \vec{E}d\vec{D} + \vec{H}d\vec{B} \end{cases}$$

各向同性 $\vec{D} = \varepsilon\vec{E}$ $\vec{B} = \mu\vec{H}$

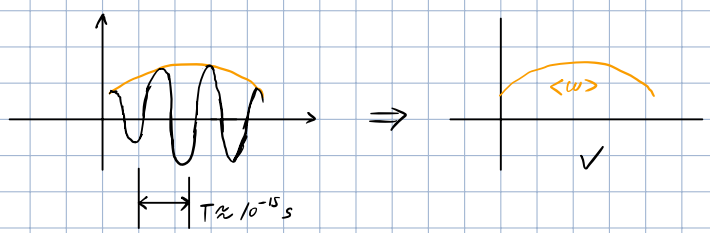
$$w = \frac{1}{2}(\varepsilon E^2 + \mu H^2) = \frac{1}{2}(\vec{E}\cdot\vec{D} + \vec{H}\cdot\vec{B})$$

平面波 $\nabla\times\vec{E} = -\frac{\partial\vec{B}}{\partial t} \Rightarrow \vec{k}\times\vec{E} = -\omega\vec{B}$ $\frac{|E|}{|B|} = v = \frac{1}{\sqrt{\mu\varepsilon}}$

$$w = \varepsilon E^2 = \frac{1}{\mu}B^2$$

测 ω 时，实际只能测到 $T \approx 10^{-15}s$，只能取平均.

$$\langle \omega \rangle = \frac{1}{T}\int_0^T \omega dt$$



对平面波

$$\vec{E} = \vec{E}_0 \cos(\omega t - \vec{k}\cdot\vec{r})$$

代入：

$$\langle \omega \rangle = \frac{1}{T}\int_0^T \frac{1}{2}\varepsilon E_0^2 \left\{1+\cos[2(\omega t - \vec{k}\cdot\vec{r})]\right\} dt$$

$$= \frac{1}{2}\varepsilon E_0^2$$

$$\vec{S} = \vec{E} \times \left(\frac{1}{\mu\omega}\vec{k}\times\vec{E}\right) = \omega \vec{v}_\phi$$

$$\langle \vec{S} \rangle = \langle \omega \rangle \vec{v}_\phi$$

$$I = \frac{1}{2}\sqrt{\frac{\varepsilon}{\mu}}E_0^2 = \frac{1}{2c\mu_r}nE_0^2 \qquad \begin{cases} \mu_r = 1 \\ \varepsilon_r = n^2 \end{cases}$$

$I \propto E_0^2$，$I \approx E_0^2$

光间色散

ε μ.

$$n(\omega) = \frac{1}{\sqrt{\varepsilon(\omega)\mu(\omega)}}$$

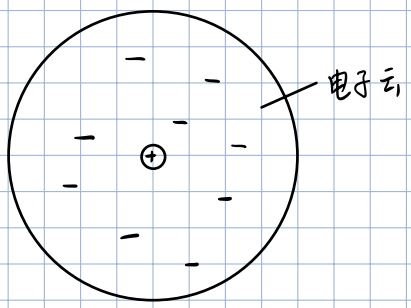
$\varepsilon(\omega) \longrightarrow$ 电

$\mu(\omega) \longrightarrow$ 磁

$\mu_r \mu_0 \approx 1$

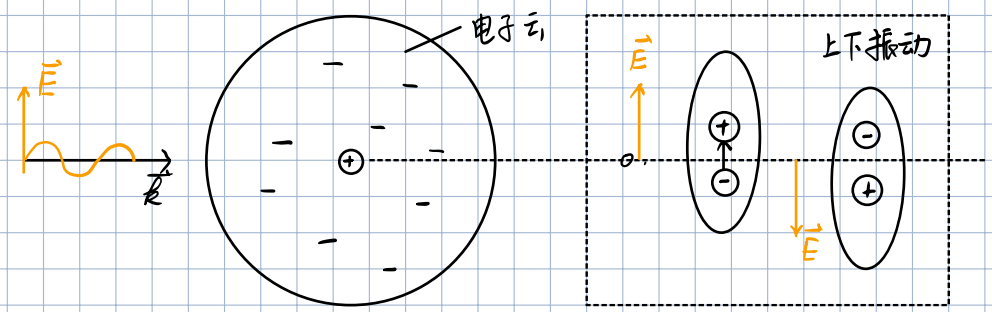
Lorentz色散模型

1. 稀薄气体.

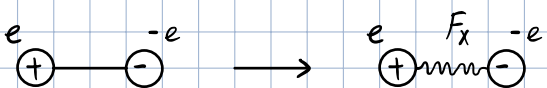


正负电核数目相同中心重合，不带电

射入光束



果冻模型 ⟶ 电偶极振子



要求力：① 有平衡位置，② 有周期性.

$$F = -k\frac{e^2}{r} \quad \times$$

对其作 Taylor 展开：

$$\vec{F}_0(x) = \underbrace{F(0)}_{=0} + \underbrace{AF'(0)x}_{\text{线性}} + \underbrace{BF''(0)x^2}_{\text{非线性}} + \cdots$$

$$F(x) = kF'(0)x \longrightarrow \text{Hook's law}$$

1. $F = -kx = -m\omega_0^2 x$ （ω_0 → 振动本征频率）

（$x = A\cos\omega t \quad x'' = -A\omega^2\cos\omega t \quad F = x''m = -m\omega^2 x$）

2. $F_{外} = -e\vec{E}(t)$

3. 阻尼 $f = -m\Gamma\vec{v}(t)$ （Γ → 阻尼系数）　$\tau = \frac{1}{\Gamma}$ 为平均碰撞时间，弛豫。

$$F + F_{外} + f = ma$$

$$\begin{cases} \vec{x}(t) \\ \vec{v}(t) = \frac{\partial}{\partial t}\vec{x}(t) \\ \vec{a} = \frac{\partial^2}{\partial t^2}\vec{x}(t) \end{cases}$$

$$m\frac{\partial^2}{\partial t^2}\vec{x}(t) + m\Gamma\frac{\partial}{\partial t}\vec{x}(t) + m\omega_0^2\vec{x}(t) = -e\vec{E}(t)$$

对 x 作 FT，$x(t) \to x(\omega)$

$$\begin{cases} x(t) = \int_{-\infty}^{\infty} x(\omega)e^{-i\omega t}d\omega \\ \frac{\partial}{\partial t}x = -i\omega x \\ \frac{\partial^2}{\partial t^2}x = -\omega^2 x \end{cases}$$

$$-\omega^2 mx - im\omega\Gamma x + m\omega_0^2 x = -e\vec{E}(\omega)$$

$$x(\omega) = -\frac{e}{m}\frac{\vec{E}(\omega)}{\omega_0^2 - \omega^2 - i\omega\Gamma}$$

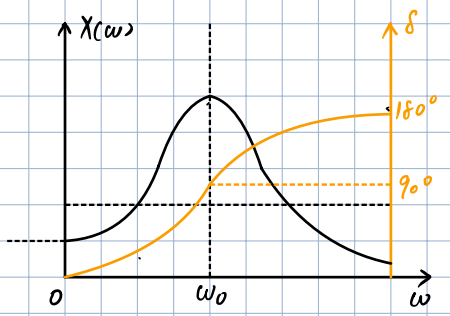
$$= -\frac{e}{m}\frac{\vec{E}(\omega)}{(\omega_0^2-\omega^2)^2 + \omega^2\Gamma^2}e^{-i(\omega t-\delta)}$$

$\tan\delta = \frac{\Gamma\omega}{\omega_0^2-\omega^2}$

$\omega \gg \omega_0$ $\quad \chi(\omega) \longrightarrow 0$ $\quad$ 几乎不振动.

$\omega \approx \omega_0$ $\quad \chi(\omega)$ 最大. $\longrightarrow$ 共振.

$\omega \ll \omega_0$ $\quad \chi \longrightarrow -\frac{e}{m}\frac{E_0}{\omega_0}$



dipole

$$\vec{\mu}(\omega) = Ne\,\vec{x}(\omega)$$

$$\vec{P}(\omega) = -\vec{\mu}(\omega) \qquad \vec{P} = \varepsilon_0 \chi \vec{E}$$

$$\vec{D}(\omega) = \varepsilon(\omega)\vec{E}(\omega) = \varepsilon(\omega)\vec{E}(\omega) + \vec{P}(\omega)$$

$$\chi(\omega) = \frac{Ne^2}{m\varepsilon_0}\frac{1}{\omega^2 - \omega_0^2 - i\omega\Gamma} \quad \left(\frac{Ne^2}{m\varepsilon_0} \equiv \omega_p^2 \text{ 等离子体频率}\right)$$

$$\boxed{\varepsilon(\omega) = \varepsilon_0 + \frac{\omega_p^2}{\omega_0^2 - \omega^2 - i\omega\Gamma} = n^2(\omega)}$$

$$\tilde{n}(\omega) = n(\omega) + iK(\omega)$$

$n(\omega)$: 色散特性 $\quad K(\omega)$: 消光系数. $\longrightarrow$ 损耗.

由折射率定义:

$$\tilde{n}^2(\omega) = \tilde{\varepsilon}_r = \frac{\tilde{\varepsilon}}{\varepsilon_0}$$

$$\tilde{n}^2(\omega) = 1 + \frac{Ne^2}{m\varepsilon_0}\frac{1}{\omega_0^2 - \omega^2 - i\Gamma\omega}$$

$$\boxed{\begin{cases} \varepsilon_r'(\omega) = n^2 - k^2 = 1 + \dfrac{Ne^2}{\varepsilon_0 m}\dfrac{\omega_0^2 - \omega^2}{(\omega_0^2 - \omega^2)^2 + \Gamma^2\omega^2} \\ \varepsilon_r''(\omega) = 2kn = \dfrac{Ne^2}{\varepsilon_0 m}\dfrac{\omega}{(\omega_0^2 - \omega^2)^2 + \Gamma^2\omega^2} \end{cases}}$$

ε_r' ⟶ 色散关系　　ε_r'' ⟶ 吸收关系.

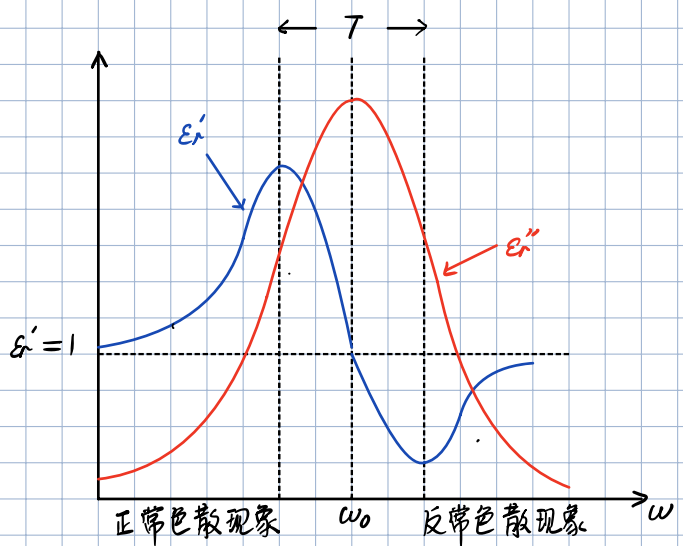
① $\omega = \omega_0$　共振吸收.　$\varepsilon_r'(\omega)$　max.

② $\omega \ll \omega_0$

$$\varepsilon_r'(\omega) = 1 + \frac{\omega_p^2}{\omega_0^2} \frac{1}{1 - 2\left(\frac{\omega}{\omega_0}\right)^2}$$

$$\varepsilon_r''(\omega) = \frac{\omega_p^2 \Gamma \omega}{\omega_0^4} \frac{1}{1 - 2\left(\frac{\omega}{\omega_0}\right)^2}$$

透明.



索姆霍兹色散方程

上述讨论中，我们假定全部电子受相同束缚作用，实际中并不相同。

∴其固有频率也不相同。

假设某情况下，固有频率为 ω_j 的电子所占比率为 ρ_j。

$$(n^2-K^2) = 1+\frac{Ne^2}{\varepsilon_0 m}\frac{\omega_0^2-\omega^2}{(\omega_0^2-\omega^2)^2+\Gamma^2\omega^2}$$

↓

$$(n^2-K^2) = 1+\frac{Ne^2}{\varepsilon_0 m}\sum_j \frac{\rho_j^2(\omega_j^2-\omega^2)}{(\omega_j^2-\omega^2)^2+\Gamma_j^2\omega^2}$$

取 $\omega=\frac{2\pi c}{\lambda}$，$\omega_j=\frac{2\pi c}{\lambda_j}$ $b_j=\frac{Ne^2}{\varepsilon_0 m}\frac{\lambda_j\rho_j}{(2\pi c)^2}$ $g_j=\frac{\Gamma_j^2\lambda_j^4}{(2\pi c)^2}$

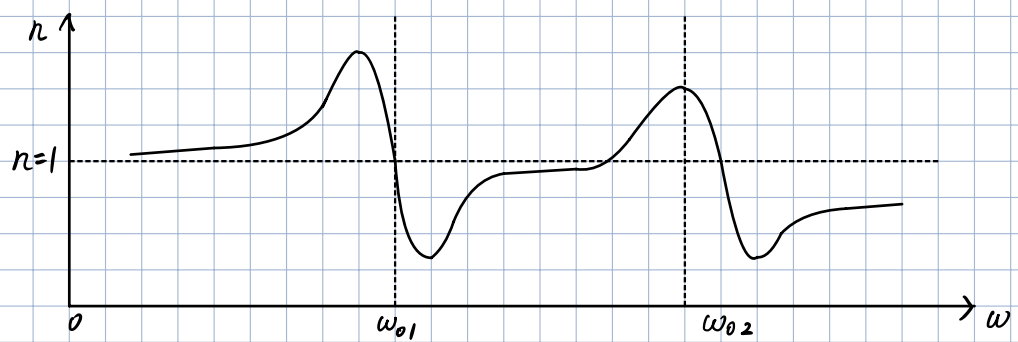
代入得：

索姆霍兹色散方程

$$n^2-K^2 = 1+\sum_j \frac{b_j\lambda^2}{(\lambda^2-\lambda_j^2)^2+\frac{g_j\lambda^2}{\lambda^2-\lambda_j^2}}$$

when 阻尼力很小，K很小，

$$n^2 = 1+\sum_j \frac{b_j\lambda^2}{(\lambda^2-\lambda_j^2)^2+g_j\lambda^2/(\lambda^2-\lambda_j^2)}$$



Sellmeier equation

在介质共振吸收区域外，

$$K=0,\quad g_j=0$$

玄姆霍兹色散方程可简化为：

$$n^2 = 1 + \sum_j \frac{b_j \lambda^2}{\lambda^2-\lambda_j^2}$$ Sellmeier equation

Cauchy equation

对给定波段，如共振波段附近，

when $\lambda \gg \lambda_0$

Sellmeier equation ⟶ Cauchy equation

$$n^2 = 1 + \frac{b\lambda^2}{\lambda^2-\lambda_0^2} = 1 + b + b\left(\frac{\lambda_0}{\lambda}\right)^2 + b\left(\frac{\lambda_0}{\lambda}\right)^4 + \cdots$$?

合并常数因子后可写为：

$$n^2 = A + \frac{B}{\lambda^2} + \frac{C}{\lambda^4} + \cdots$$ Cauchy equation

群速度与相速度

以平面波为例.

$$E_1(x,t) = A_0 e^{-i(\omega_1 t - k_1 x)}$$

$$E_2(x,t) = A_0 e^{-i(\omega_2 t - k_2 x)}$$

$$E(x,t) = E_1 + E_2 = 2A_0 \cos\left(\underbrace{\frac{\omega_2-\omega_1}{2}}_{\Delta\omega} t - \underbrace{\frac{k_2-k_1}{2}}_{\Delta k} x\right) e^{-i\left(\underbrace{\frac{\omega_2+\omega_1}{2}}_{\bar{\omega}} t - \underbrace{\frac{k_2+k_1}{2}}_{\bar{k}} x\right)}$$

群速度 $v_p = \frac{\bar{\omega}}{\bar{k}}$

相速度 $v_g = \frac{\Delta\omega}{\Delta k} \to \frac{d\omega}{dk}$

$$v_p = \frac{\omega}{k} \Longrightarrow \omega = k v_p = \frac{2\pi}{\lambda}\frac{c}{n}$$

$$v_g = \frac{d\omega}{dk} = \frac{d(kv_p)}{dk} = v_p + \frac{k\,dv_p}{dk} = v_p - \lambda\frac{dv_p}{d\lambda} = \frac{c}{n}\left(1 + \frac{\lambda}{n}\cdot\frac{dn}{d\lambda}\right)$$

由 Cauchy 公式:

$$n^2 = A + \frac{B}{\lambda^2} + \cdots$$

ε' ⟶ 色散关系　　ε'' ⟶ 吸收关系.

K-K关系. ⟶ 因果律.

$$\varepsilon'(\omega) = 1 + \frac{2}{\pi} P \int_0^{\infty} \frac{\omega' \varepsilon''(\omega')}{\omega'^2 - \omega^2} d\omega'$$

$$\varepsilon''(\omega) = -\frac{2\omega}{\pi} P \int_0^{\infty} \frac{\varepsilon'(\omega) - 1}{\omega'^2 - \omega^2} d\omega'$$

$$n(\omega) = 1 + \frac{2}{\pi} P \int_0^{\infty} \frac{\omega' k(\omega') - \omega k(\omega)}{\omega'^2 - \omega^2} d\omega'$$

$$k(\omega) = -\frac{2\omega}{\pi} P \int_0^{\infty} \frac{n(\omega')}{\omega'^2 - \omega^2} d\omega'$$

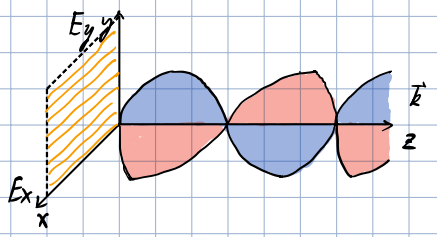
偏振

$$\vec{E}(\vec{r},t) = \vec{E_0} e^{-i(\omega t - \vec{k}\cdot\vec{r} - \varphi_0)}$$

$$\vec{E_0} = \vec{E}(x,y,z) = E_x \hat{x}_0 + E_y \hat{y}_0 + E_z \hat{z}_0$$

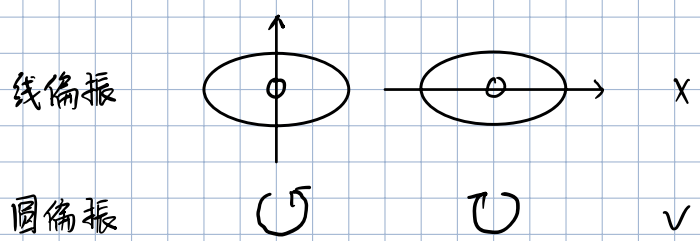
假设传输方向 $\vec{k}$ 沿 z 方向　　$\vec{E_z} = 0$　　$\vec{E_0} = \vec{E}(x,y)$

偏振



偏振应用:

3D 眼镜.　　线 ξ or 圆 ξ ?



偏振描述

$$\begin{cases} E_x = A_x \cos(\omega t - \phi_x) \\ E_y = A_y \cos(\omega t - \phi_y) \end{cases}$$

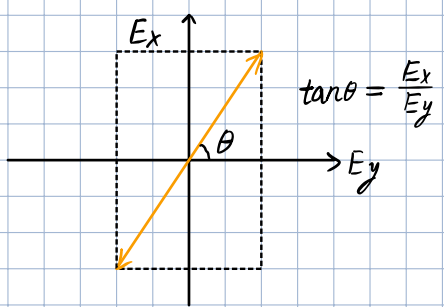
证明：$\delta = \phi_x + \omega t - \phi_y - \omega t$ 展开

$$\left(\frac{E_x}{A_x}\right)^2 + \left(\frac{E_y}{A_y}\right)^2 - 2\frac{E_x E_y}{A_x A_y}\cos\delta = \sin^2\delta .$$

$$\delta = \phi_x - \phi_y$$

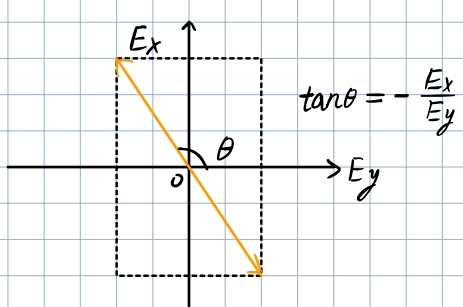
for $\delta = 0$

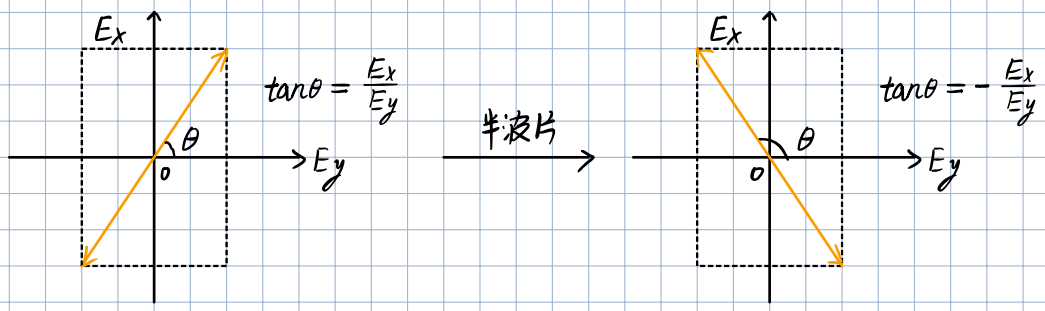
$$\frac{E_x}{A_x} = \frac{E_y}{A_y}$$



for $\delta = \pi$.

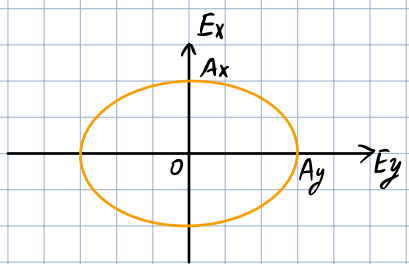
$$\frac{E_x}{A_x} + \frac{E_y}{A_y} = 0$$



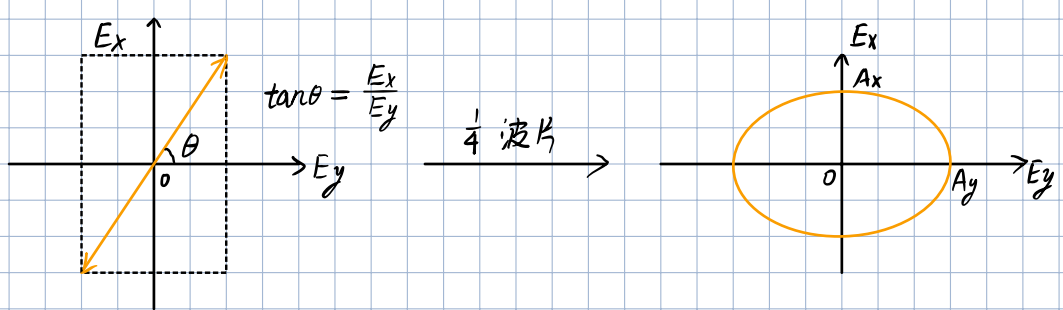


for $\delta = \pm \frac{\pi}{2}$.

$$\frac{E_x^2}{A_x^2} + \frac{E_y^2}{A_y^2} = 1$$



when $A_x = A_y$, 圆.　　when $A_x \neq A_y$, 椭圆.



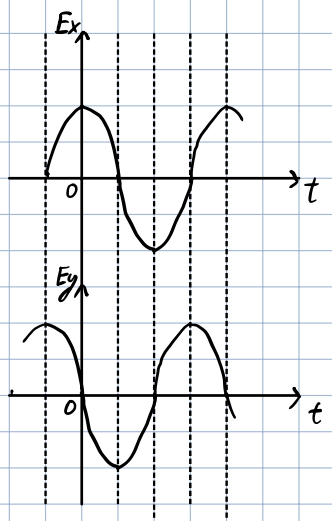
$\delta = ?$ 左旋 or 右旋?

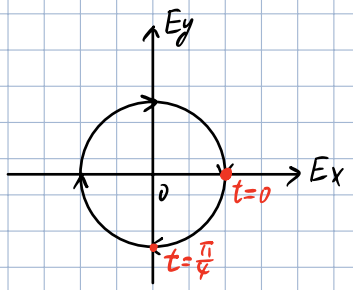
$\delta = \phi_x - \phi_y = \frac{\pi}{2}$ or $-\frac{\pi}{2}$ $(\phi_x = 0 \quad \phi_y = \frac{\pi}{2})$

$E_x = A_x \cos(\omega t - \phi_x) = A_x \cos \omega t$

$E_y = A_y \cos(\omega t - \phi_y) = A_y \cos \omega t$

for $\delta = -\frac{\pi}{2}$





for $\delta = \frac{\pi}{2}$, 右旋

for $\delta = -\frac{\pi}{2}$, ...

dim 偏振度.

$$P=\frac{I_{max}-I_{min}}{I_{max}+I_{min}}$$

完全偏振光.

$P=0$ 自然光. ⟶ { δ 恒定; δ 随机; 空间分布均匀.

$0<P<1$ 部份偏振光.

$P=1$ 线偏振光